RANK-FEEDBACK SPLITTING OF FINITE ABELIAN p-GROUPS: A SHARP TRIANGULAR CLOCK AND EXACT TARGET FIBRES

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Abstract. For a finite abelian p-group G, let $d(G)$ be its minimum number of generators and iterate

$$\mathsf{F}(G) = p^{d(G)}G \oplus G[p^{d(G)}].$$

If G has partition type $\lambda = (a_1, \dots, a_r)$, the induced rule keeps each $a_i \le r$ and splits each $a_i > r$ into $(r, a_i - r)$. We prove that the recurrent types are exactly the fixed types $a_1 \le r$ and give their ordinary generating function. More sharply, for every type of order p^n and initial rank r_0, a transient of length d satisfies $n \ge r_0(d+1) + \binom{d}{2}$. Consequently the maximum transient length is

$$\left\lceil \frac{\sqrt{8n+1}-3}{2} \right\rceil,$$

and the unique type attaining it is the cyclic type (n), whose whole orbit is explicit. Finally, a bounded-coefficient formula counts the one-step fibre over every target partition and yields a necessary and sufficient image criterion. The classification of finite abelian groups, cyclic kernel/image formulas, torsion terminology, and Gaussian-binomial partition enumeration are assigned zero contribution credit. Exact enumeration is used only for falsification, and external release remains on hold.

1. The literal operator and complete statement

Fix a prime p. Let $\mathcal{A}_{p,n}$ be the set of isomorphism classes of finite abelian groups of order p^n. For $G \in \mathcal{A}_{p,n}$ put

$$d(G) = \dim_{\mathbb{F}_p}(G/pG), \qquad G[p^s] = \{g \in G : p^s g = 0\},$$

and define

$$\mathsf{F}(G) = p^{d(G)}G \oplus G[p^{d(G)}]. \tag{1}$$

The direct sum in (1) is external. The zero group is fixed by convention.

Write $\mathcal{P}_n$ for the integer partitions of n. Parts are decreasing, $\ell(\lambda)$ is their number, and $m_j(\lambda)$ is the multiplicity of j. Put $\mathrm{T}_t = t(t+1)/2$. The *entry time* $\tau(\lambda)$ is the least $t \ge 0$ for which the tth iterate is recurrent.

For later use, fix a target $\mu \in \mathcal{P}_n$ of length L and write $m_j = m_j(\mu)$. For

$$\left\lceil \frac{L}{2} \right\rceil \le r \le L, \qquad c_r = L - r,$$

if $m_r \ge c_r$, define

$$q_j^{(r)} = m_j - c_r \mathbf{1}_{\{j=r\}}, \qquad h_r = \sum_{j>r} q_j^{(r)}, \tag{2}$$

and set

$$A_r(\mu) = \begin{cases} [u^{c_r - h_r}] \displaystyle\prod_{j=1}^{r} (1 + u + \cdots + u^{q_j^{(r)}}), & m_r \ge c_r,\ h_r \le c_r, \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

Theorem 1.1 (Dynamics, sharp clock, and inverse geometry)**.** *For every $n \ge 1$, the following statements hold.*

(i) *Under the usual type bijection*

$$\lambda = (a_1, \dots, a_r) \longleftrightarrow G_\lambda = \bigoplus_{i=1}^{r} C_{p^{a_i}},$$

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the group operator (1) *is the weight-preserving partition map*

$$\mathsf{F}(\lambda) = \mathrm{sort}\left(\biguplus_{a_i \le r} \{a_i\} \uplus \biguplus_{a_i > r} \{r, a_i - r\} \right). \tag{4}$$

In particular, its type dynamics is independent of p.

(ii) *A type* $\lambda = (a_1, \ldots, a_r)$ *is recurrent if and only if it is fixed, if and only if* $a_1 \le r$. *If* f_n *counts fixed types and* $f_0 = 1$, *then, as a formal power series,*

$$\sum_{n \ge 0} f_n z^n = 1 + \sum_{r \ge 1} z^r \begin{bmatrix} 2r-1 \\ r \end{bmatrix}_z. \tag{5}$$

(iii) *If* $\lambda \vdash n$ *has initial length* r_0 *and entry time* d, *then*

$$n \ge r_0(d+1) + \binom{d}{2}. \tag{6}$$

The exact maximum and its unique maximizer are

$$\max_{\lambda \vdash n} \tau(\lambda) = D(n) := \left\lceil \frac{\sqrt{8n+1} - 3}{2} \right\rceil, \qquad \tau(\lambda) = D(n) \Longleftrightarrow \lambda = (n). \tag{7}$$

Moreover, for $0 \le t \le D(n)$,

$$\mathsf{F}^t((n)) = \mathrm{sort}(n - \mathrm{T}_t, t, t-1, \ldots, 1), \tag{8}$$

where the list after $n - \mathrm{T}_t$ *is empty when* $t = 0$.

(iv) *The fibre over every target, including a target outside the image, is*

$$|\mathsf{F}^{-1}(\mu)| = \sum_{r = \lceil L/2 \rceil}^{L} A_r(\mu). \tag{9}$$

The corresponding exact image criterion is

$$\mu \in \operatorname{im} \mathsf{F} \quad \Longleftrightarrow \quad \textit{some } r \in [\lceil L/2 \rceil, L] \textit{ satisfies } m_r \ge L - r \textit{ and } \sum_{j > r} m_j \le L - r. \tag{10}$$

2. From groups to a feedback split

The classification and cyclic decomposition of finite abelian groups are standard; see, for example, Fuchs [5]. We nevertheless include the short calculation that fixes the literal map and all conventions.

Proposition 2.1 (Group/type identity)**.** *The operator* (1) *preserves group order and induces* (4).

Proof. For $G_\lambda = \bigoplus_{i=1}^r C_{p^{a_i}}$ one has $G_\lambda / pG_\lambda \cong \mathbb{F}_p^r$, and r generators visibly suffice. Hence $d(G_\lambda) = r$. On one cyclic factor,

$$p^r C_{p^a} \cong C_{p^{\max(a-r,0)}}, \qquad\qquad C_{p^a}[p^r] \cong C_{p^{\min(a,r)}}, \tag{11}$$

with an exponent zero denoting the trivial factor. Thus $a \le r$ yields one exponent a, while $a > r$ yields exponents r and $a - r$. Direct sums commute with both constructions, proving (4).

Multiplication by p^r also gives an exact sequence

$$0 \longrightarrow G[p^r] \longrightarrow G \longrightarrow p^r G \longrightarrow 0.$$

Therefore $|p^r G|\,|G[p^r]| = |G|$, so (1) indeed remains in $\mathcal{A}_{p,n}$. □

Both (11) and the order identity are classical inputs, not contribution claims. The dynamics begins only when their exponent r is fed back from the current state.

3. Fixed and recurrent types

For a type λ of length r, set

$$c(\lambda) = |\{i : a_i > r\}|. \tag{12}$$

Every such part splits into two and every other part remains one part, so

$$\ell(\mathsf{F}(\lambda)) = r + c(\lambda). \tag{13}$$

Proposition 3.1 (No nontrivial recurrence)**.** *A partition is recurrent precisely when* $c(\lambda) = 0$, *equivalently when* $a_1 \le \ell(\lambda)$, *and all recurrent types are fixed.*

Proof. If $c(\lambda) = 0$, (4) leaves every part unchanged. If $c(\lambda) > 0$, (13) strictly increases the length. A partition of n has length at most n, so every orbit eventually reaches a fixed type. Strict length increase also prevents a nonfixed state from lying on a directed cycle. □

Proof of (5). Fix the length r. A fixed type has exactly r positive parts, all at most r. Subtracting one from every part produces a Ferrers diagram in an r-by-$(r-1)$ rectangle, with zero rows allowed. Its size enumerator is the Gaussian polynomial $\left[{2r-1 \atop r}\right]_z$. Restoring the removed column shifts the weight by r and gives the rth summand of (5). The initial 1 is the empty type. Rectangle enumeration by Gaussian polynomials is standard partition theory [1] and receives zero credit here. □

4. Frozen markers and the unique sharp clock

Let

$$\lambda^{(0)} \longmapsto \lambda^{(1)} \longmapsto \cdots \longmapsto \lambda^{(d)}$$

be an orbit segment in which the first d states are nonfixed and the last is fixed. Write $r_t = \ell(\lambda^{(t)})$ and $c_t = c(\lambda^{(t)})$ for $0 \le t < d$.

Lemma 4.1 (Pointwise marker budget)**.** *For every such segment,*

$$n \ge r_0 + \sum_{t=0}^{d-1} c_t r_t \ge r_0(d+1) + \binom{d}{2}. \tag{14}$$

Proof. Temporarily tag the r_0 initial parts. Within each tag, distinguish one positive *residual*. When a residual part $a > r_t$ splits, designate the new part r_t as a marker and retain $a - r_t > 0$ as that tag's residual. Unsplit parts remain residuals. The map never merges parts.

Equation (13) gives $r_{t+1} = r_t + c_t > r_t$. Hence a marker of size r_t is at most every later rank and can never split later. At time d there are consequently c_t disjoint permanent markers of weight r_t from each transition t, plus one positive residual for every initial tag. Their weights give the first inequality in (14).

Since every $c_t \ge 1$, the rank recurrence implies $r_t \ge r_0 + t$. Therefore

$$r_0 + \sum_{t=0}^{d-1} c_t r_t \ge r_0 + \sum_{t=0}^{d-1} (r_0 + t) = r_0(d+1) + \binom{d}{2},$$

which proves the second inequality. □

Theorem 4.2 (Sharpness and uniqueness)**.** *Equations* (6)–(8) *hold.*

Proof. Taking $r_0 \ge 1$ in Lemma 4.1 gives

$$n \ge \mathrm{T}_d + 1,$$

so $\mathrm{T}_d < n$. The largest integer with that strict inequality is $D(n)$ in (7); equivalently $\mathrm{T}_{D(n)} < n \le \mathrm{T}_{D(n)+1}$. This proves the universal upper bound and the displayed radical form.

Starting from (n), suppose $0 \le t < D(n)$ and (8) holds. Its length is $t+1$, its parts $t, t-1, \ldots, 1$ are already at most that length, and

$$n - \mathrm{T}_t > t + 1$$

because $\mathrm{T}_{t+1} < n$. Thus only $n - \mathrm{T}_t$ splits, into $(t+1, n - \mathrm{T}_{t+1})$, proving the next instance of (8). At $t = D(n)$, the reverse inequality $n \le \mathrm{T}_{D(n)+1}$ says that $n - \mathrm{T}_{D(n)} \le D(n) + 1$; all parts are then at most the rank, so this state is fixed. Hence (n) attains the bound.

Finally, suppose a type of length $r_0 \ge 2$ had entry time $D = D(n)$. The pointwise bound would give

$$n \ge 2(D+1) + \binom{D}{2} = \mathrm{T}_{D+1} + 1,$$

contrary to $n \le \mathrm{T}_{D+1}$. Thus every maximizer has $r_0 = 1$, and the only partition of n with one part is (n). This also covers $n = 1$, for which $D(n) = 0$. □

5. Every-target fibres and the image

Theorem 5.1 (Marker removal and bounded choices)**.** *The fibre formula* (9) *and image criterion* (10) *hold for every $\mu \vdash n$.*

Proof. Fix a source rank r, and let c of its r parts exceed r. Each of those c parts contributes a marker r and one positive remainder; every other source part is unchanged. Hence the target length is $L = r + c$. Since $0 \le c \le r$, the possible source ranks are exactly in $[\lceil L/2 \rceil, L]$, with $c = c_r = L - r$.

For one such r, the target must contain at least c_r copies of r. Remove those marker copies. The residual multiset has counts $q_j^{(r)}$ and exactly r parts. Every residual part greater than r must have come from a split source part, so all h_r such copies are forced remainders. Among the copies of sizes $1, \dots, r$, choose another $c_r - h_r$ remainders. If s_j copies of size j are chosen, then

$$0 \le s_j \le q_j^{(r)}, \qquad \sum_{j=1}^{r} s_j = c_r - h_r.$$

The number of such multiplicity vectors is exactly the coefficient in (3).

The reconstruction is explicit: combine every selected remainder j with one removed marker to make the source part $r + j > r$, and retain each unselected residual part as an unsplit source part. It produces c_r large and $r - c_r$ small source parts, hence a source of rank r. The construction and marker removal are inverse, and different source ranks are disjoint. Summing over r proves (9).

It remains to read positivity. If $m_r \ge c_r$ and $h_r \le c_r$, then there are $r - h_r$ residual copies of size at most r, and

$$0 \le c_r - h_r \le r - h_r$$

because $c_r \le r$. Thus a choice exists, so $A_r(\mu) > 0$. Conversely, a preimage necessarily supplies both the c_r markers and at most c_r large residuals. Since $h_r = \sum_{j>r} m_j$, these are precisely the two conditions in (10). □

6. Ownership boundary and exact controls

All classical structure is subtracted. The finite abelian p-group classification, $d(G) = \dim_{\mathbb{F}_p} G/pG$, and the cyclic formulas (11) are standard group theory [5]. Delaunay and Jouhet use partition-indexed finite abelian groups and p^ℓ-torsion statistics in a different probabilistic and combinatorial setting [3]; that torsion machinery is prior art, not part of the residual claim. Ferrers diagrams, Gaussian polynomials, and formal coefficient extraction are likewise standard [1]. Existing iterated maps on partitions include multiplicity-description and continued-fraction-type systems [2, 4]. They do not supply ownership clearance for the present map, and their general partition-dynamics language receives zero credit.

The residual package is limited to the conjunction for the literal state-dependent operator (1): the monotone feedback split, pointwise marker budget, unique sharp triangular clock, and all-target one-step inverse decoder. Internally, this is not P126's synchronous balanced refinement of ordered compositions: there the threshold is fixed, the split is nearly halving, and the main result concerns complete iterate kernels. It is not P135's derived-centralizer multiplicity rule, which has mergers and genuine two-cycles, and it is not P115's linear Cartier/Frobenius coefficient dynamics. Merely sharing splitting, partition, or finite algebraic vocabulary transfers no theorem.

The paper-local exact audit constructs literal cyclic kernels and images, enumerates every partition through weight 50, compares the fixed counts with an independent Gaussian recurrence, and checks (9) and (10) over every target through weight 35. A bounded source audit did not locate the exact literal operator or theorem conjunction. This non-hit is not novelty, priority, or authorship evidence; external status is `HOLD_EXTERNAL`.

control	exact count
partition states, all $n \le 50$	1,295,970
target fibre cells, all $n \le 35$	81,155
zero-fibre targets among those cells	30,923
literal cyclic cells / base elements	176 / 10,350
fixed-OGF coefficients	51
exact assertions	18,504,770

Table 1. Dependency-free exact controls. Enumeration is falsification evidence only, not an all-weight proof or an ownership certificate.

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